

IMPERIAL

CIVE70007 GEOTECHNICAL HAZARDS · LECTURE 01

Framework for Seismic Loading

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Where this lecture fits

L01
Framework

L02-03
Seismic sources

L04-06
Rupture rates

L07-09
Ground motions

L10-12
Hazard analysis &
uncertainty

Aims & Objectives

The primary aims for this lecture are:

- 1 Connect the risk framework from the introduction to the way design loads are set
- 2 Define the end goal of hazard analysis: the hazard curve, $\lambda(IM > im)$
- 3 Set out the framework for the rest of the module

Intended learning outcomes

Following this lecture you should be able to:

- 1 Explain how a target reliability, or a minimum expected cost, becomes a design level of intensity
- 2 Perform simple calculations relating probabilities of exceedance, return periods and design lives using the Poisson distribution
- 3 Describe how optimal design levels are identified, and the trade-off between construction cost and the expected cost of failure
- 4 Identify the components of a seismic hazard analysis, and where each is covered in this module

From risk to design loads

- The introduction built risk from hazard, vulnerability and exposure:
$$\lambda(DV > dv) = \iiint G(dv \mid dm) f(dm \mid edp) f(edp \mid im) |d\lambda(IM > im)|$$
- Design practice usually asks a narrower question: *what level of intensity should this structure be designed to resist?*
- That is a special case of the same framework. With a lognormal capacity (θ, β) and a hazard curve of slope k :

$$\lambda_C = \lambda_{IM}(\theta) \exp\left(\frac{1}{2} k^2 \beta^2\right) \xrightarrow{\beta \rightarrow 0} \lambda_{IM}(im^*)$$

- If uncertainty in response and capacity is set aside, failure means “the design intensity im^* is exceeded”, so **a target failure rate becomes a target rate of exceedance on the hazard curve**
- Design codes put the neglected uncertainty back through partial factors and conservative capacities
- As in the introduction, the IM depends on the problem: codes use $S_a(T)$ at the structure’s period; here we use PGA for illustration

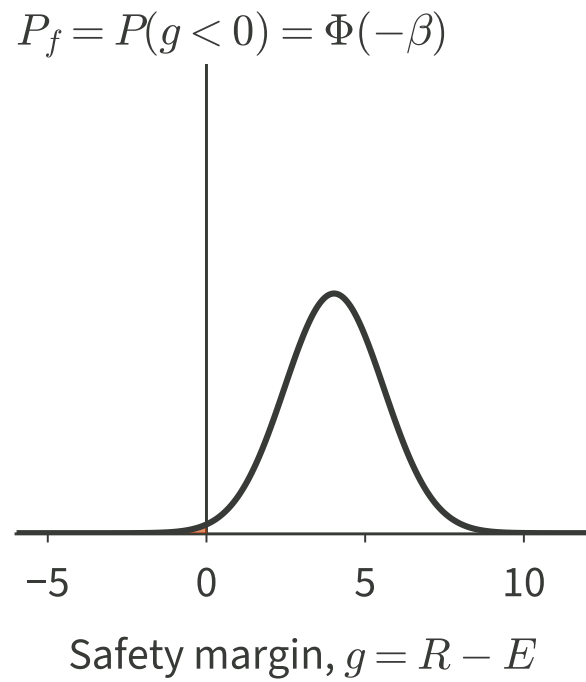
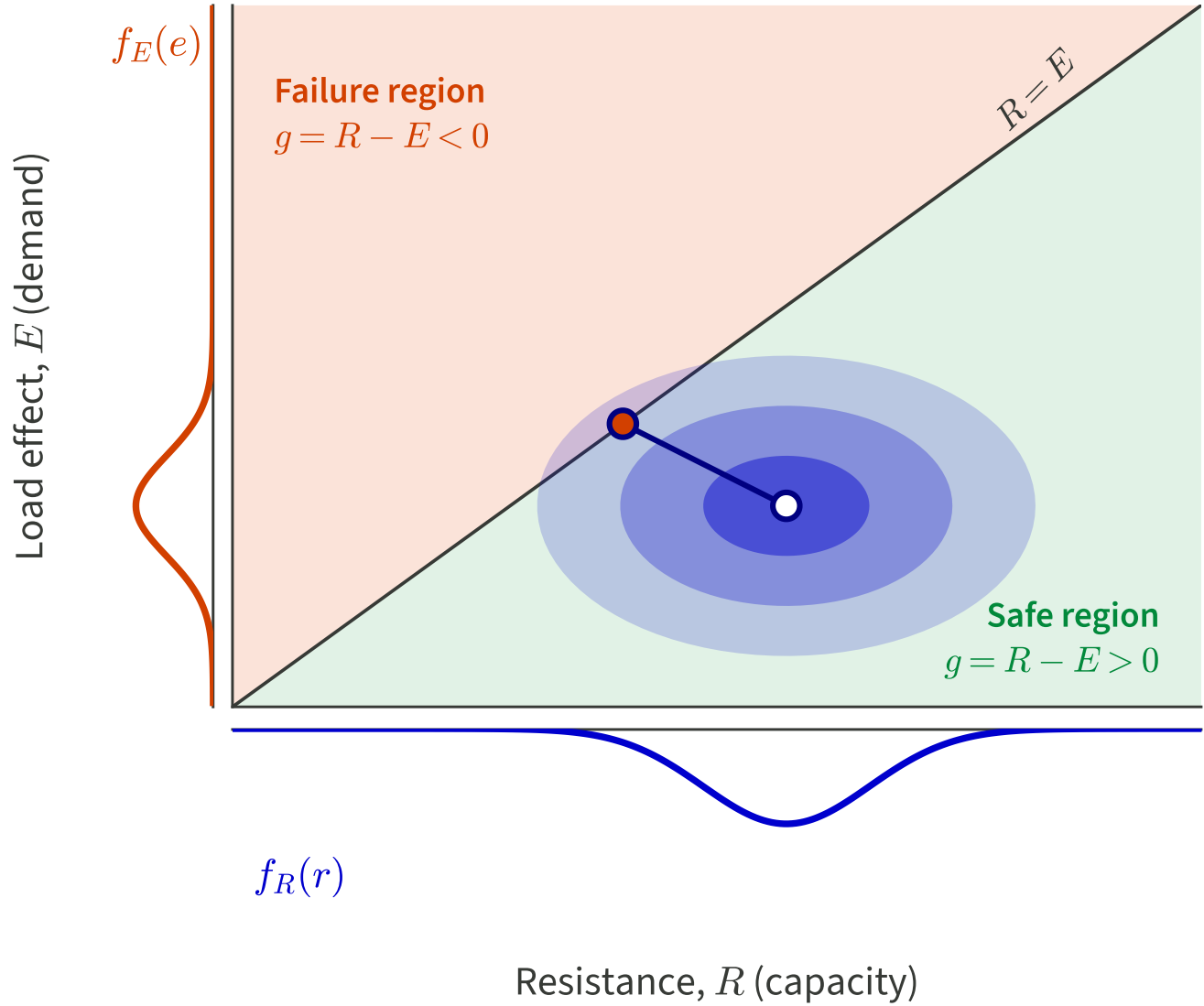
This lecture

- 1 **Reliability:** how load effects and resistance combine into a probability of failure
- 2 **Economics:** why there is an optimal level of safety, and what sets it
- 3 **Design life:** turning a target probability over a design life into a design intensity, using the hazard curve
- 4 **Hazard analysis:** what we need in order to compute that hazard curve, which is the subject of the rest of the module

All hazard curves, tables and interactive figures in this lecture come from one toy PSHA (a single fault; see “Components of seismic hazard analysis”), so the numbers are consistent throughout.

Reliability: load effects and resistance

Load effect versus resistance



- Failure when the safety margin

$$g = R - E < 0$$

- For independent normal R and E :

$$\beta = \frac{\mu_R - \mu_E}{\sqrt{\sigma_R^2 + \sigma_E^2}}, \quad P_f = \Phi(-\beta)$$

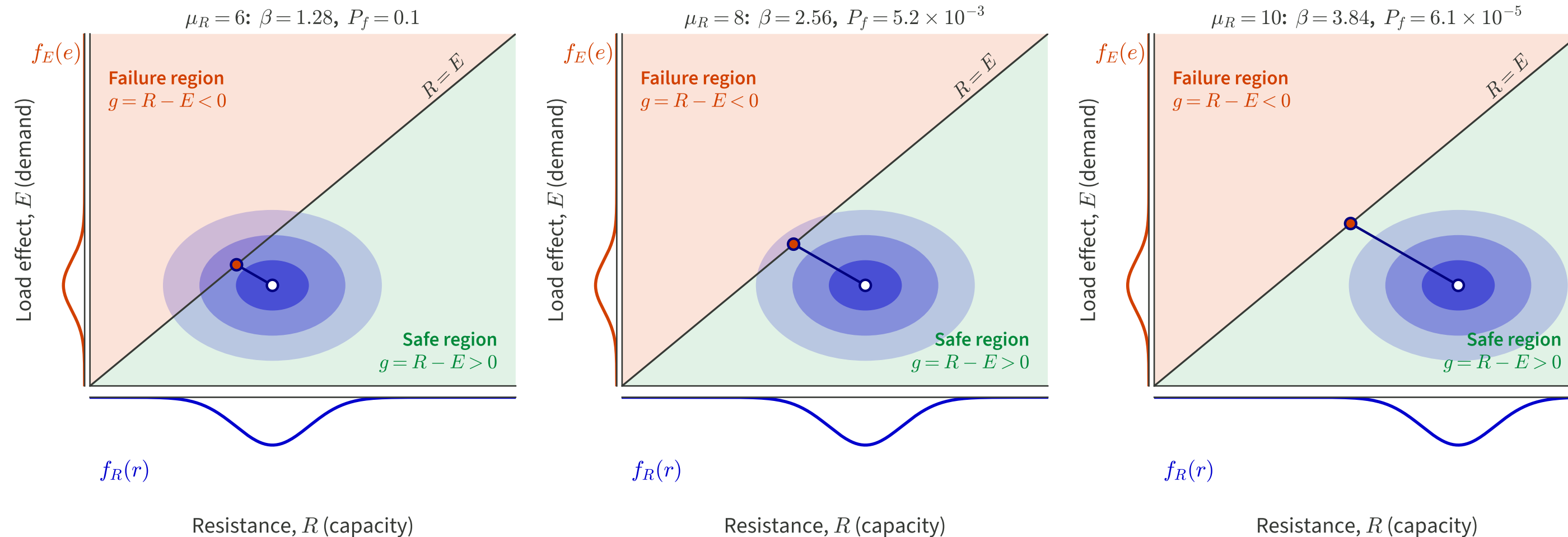
μ_R	<input type="range" value="8.0"/>	8.0
σ_R	<input type="range" value="1.20"/>	1.20
μ_E	<input type="range" value="4.0"/>	4.0
σ_E	<input type="range" value="1.00"/>	1.00

Reliability index $\beta = 2.56$

Probability of failure $P_f = 5.2 \times 10^{-3}$

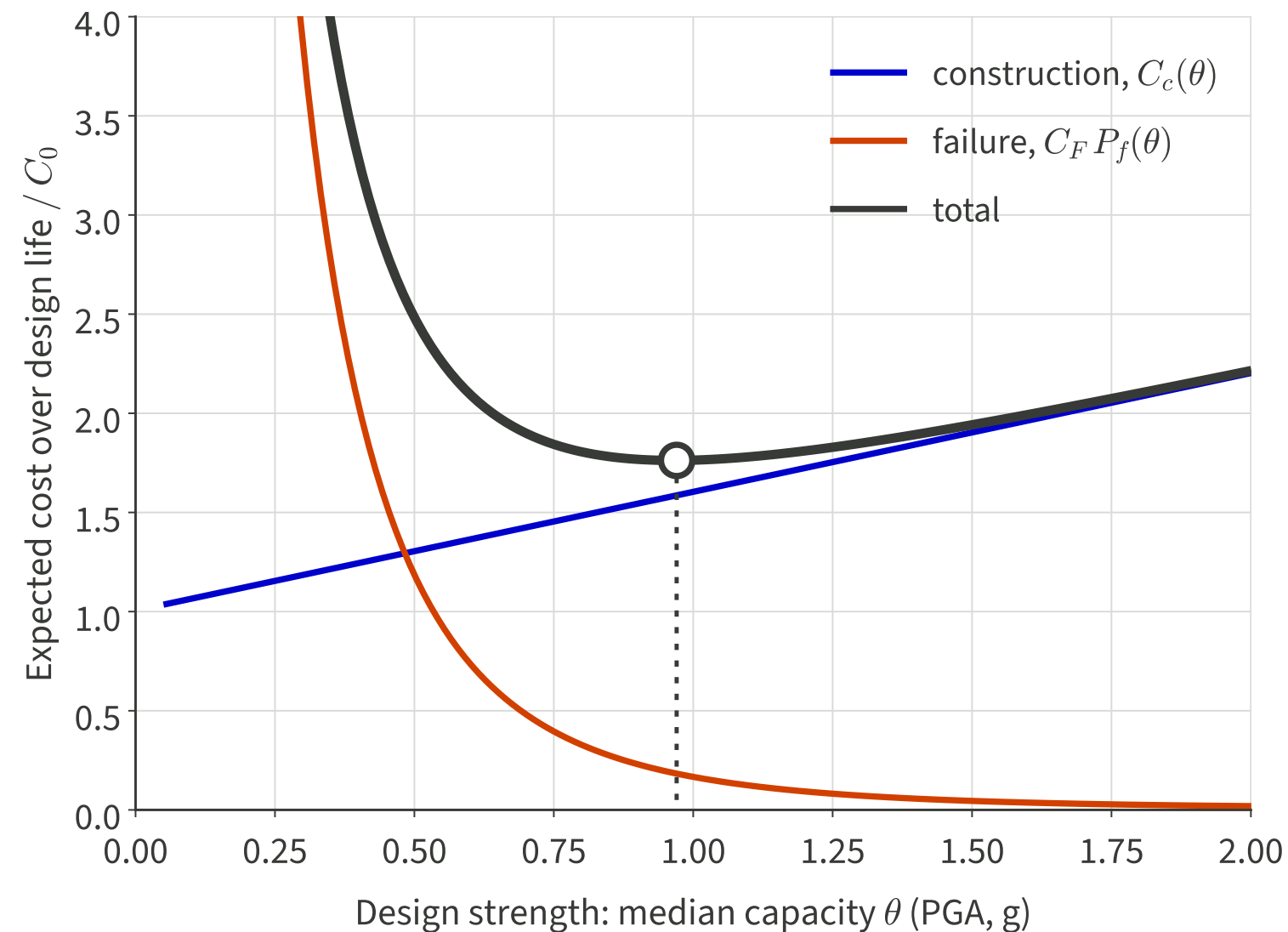
Orange point: the *design point*, the most likely combination of R and E at failure. β is the distance from the mean (white) to the design point, measured in standard deviations.

Moving the resistance



- Increasing the mean resistance moves the joint distribution away from the limit state $R = E$: β grows linearly, but $P_f = \Phi(-\beta)$ falls by orders of magnitude
- Reducing the scatter (σ_R, σ_E) has the same effect: reliability depends on the margin *relative to the uncertainty*
- For seismic loading, E is controlled by the ground-motion intensity, and its distribution comes from the hazard curve

Optimal design level



- Construction cost: $C_c = C_0(1 + 0.6 \theta)$
- Expected failure cost: $C_F P_f(\theta)$, with P_f over the design life from the hazard curve and a lognormal fragility (introduction)

C_F / C_0 30

β 0.40

T_{DL} (yr) 50

Optimal median capacity $\theta^* = 0.97$ g

P_f over the design life = 5.9×10^{-3}

Return period of failure = 8,472 yr

Return period of θ^* (hazard) = 21,082 yr

Reading the optimum

- Too little strength: construction is cheap, but the expected cost of failure dominates. Too much: failures are avoided at excessive construction cost
- The optimum moves to **stronger** designs when the consequence of failure C_F is larger, and when the capacity is more uncertain (β)
- Society also sets a **limit of tolerable failure** (life safety): any design on the safe side of it is acceptable, and the optimum is only used if it satisfies that limit
- The optimal median capacity has a return period of tens of thousands of years, far rarer than the 475-year design motion: codes design for a frequent motion and rely on safety factors and ductility to provide the margin between im^* and θ
- Both the target and the optimum are expressed as **rates**; converting them into a design intensity needs the hazard curve

From target probability to design intensity

Relationship to design life

- The probability of failure, and hence the expected cost of failure, must be tied to the design life, T_{DL}
- This link is traditionally made by assuming that exceedances of a level im follow a **Poisson process** with average annual rate $\lambda(IM > im)$
- The number of exceedances, n , in the design life then has the distribution

$$P(n) = \frac{[\lambda(IM > im) T_{DL}]^n e^{-\lambda(IM > im) T_{DL}}}{n!}, \quad E[n] = \lambda(IM > im) T_{DL}$$

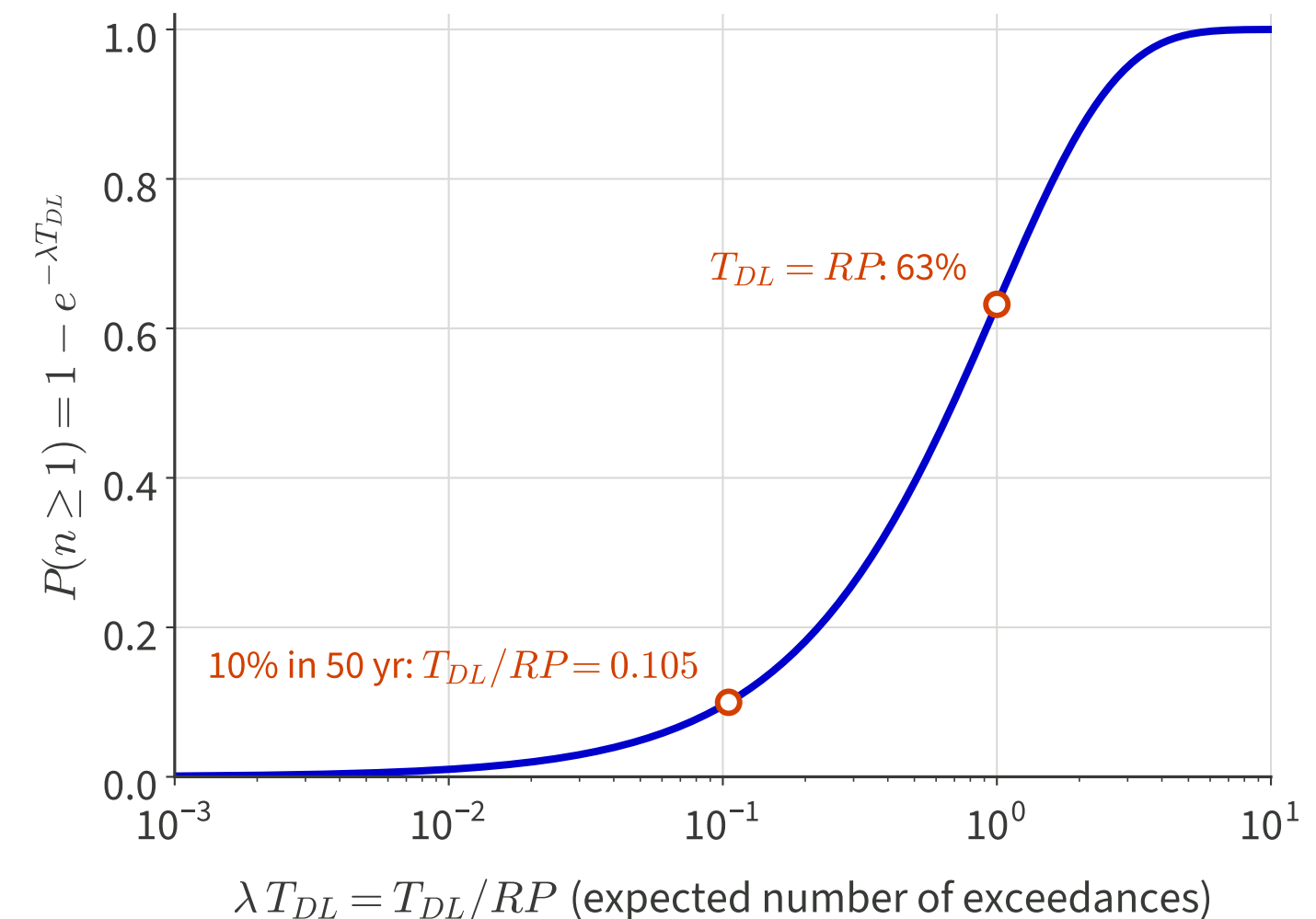
- The framework does not depend on the Poisson assumption: time-dependent models (for example, after a large earthquake) forecast hazard for a particular time window instead

Probability of at least one exceedance

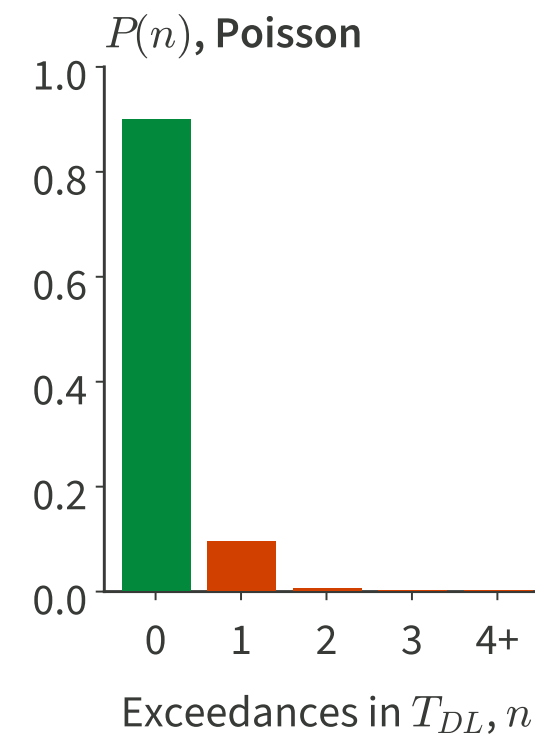
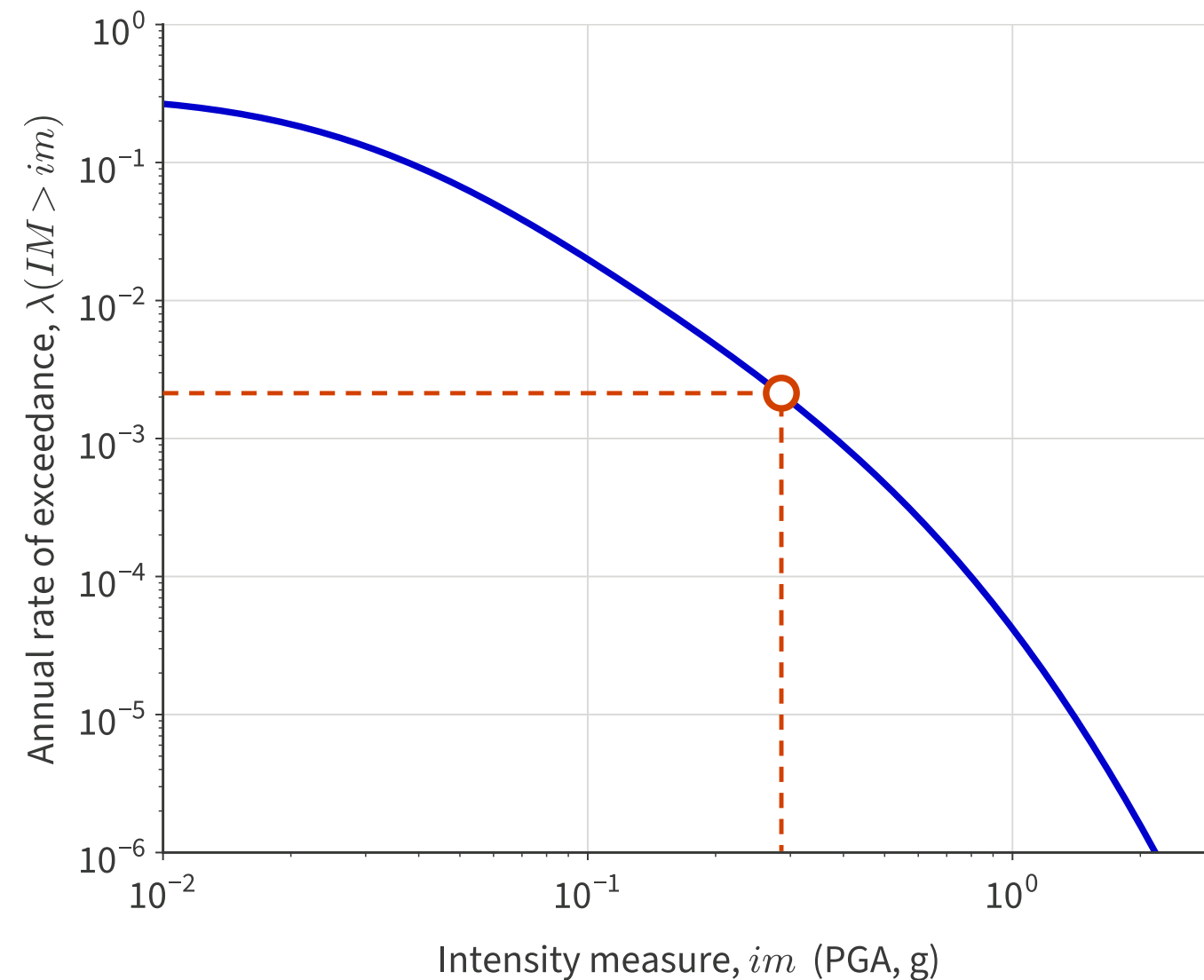
- If im is the design level, any exceedance counts as failure
- Since $\sum_n P(n) = 1$, the probability of at least one exceedance is

$$P(n \geq 1) = 1 - P(n = 0) = 1 - e^{-\lambda(IM > im) T_{DL}}$$

- Everything depends on the product λT_{DL} , the expected number of exceedances
- 10% in 50 years means $\lambda T_{DL} = 0.105$, so
 $\lambda = 2.1 \times 10^{-3}/\text{yr}$: a 475-year return period
- Over a period equal to the return period, the probability of exceedance is 63%, not 100%



From a target to a design intensity



$$\lambda^* = -\frac{\ln [1 - P(n \geq 1)]}{T_{DL}}$$

T_{DL} (yr)

$P(n \geq 1)$

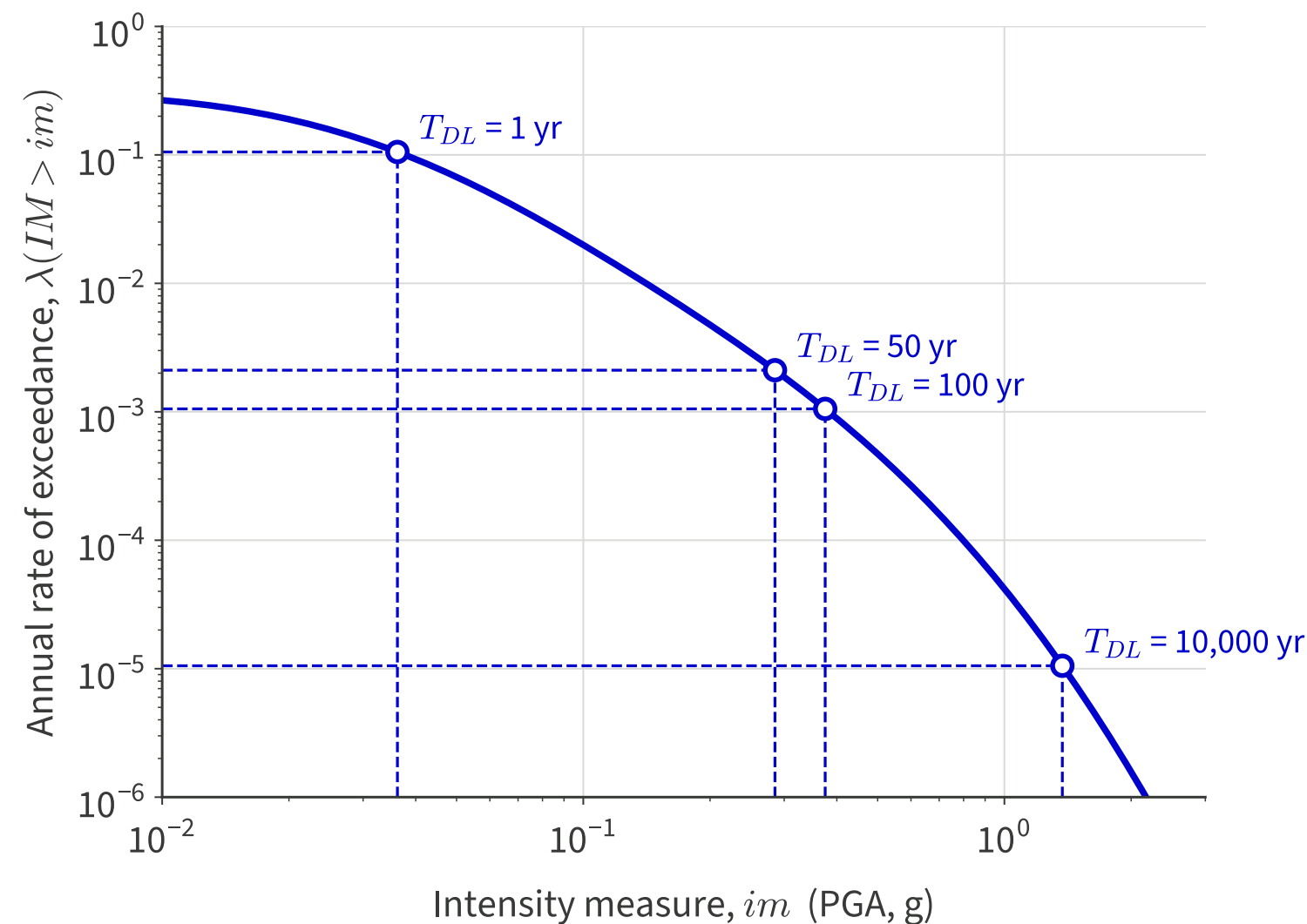
$\lambda^* = \lambda(IM > im^*) = 2.11 \times 10^{-3} / \text{yr}$

Return period = 475 yr

Design intensity $im^* = 0.29$ g

Read across from the target rate to the hazard curve, then down to the design intensity. The bars show the Poisson distribution of the number of exceedances in the design life.

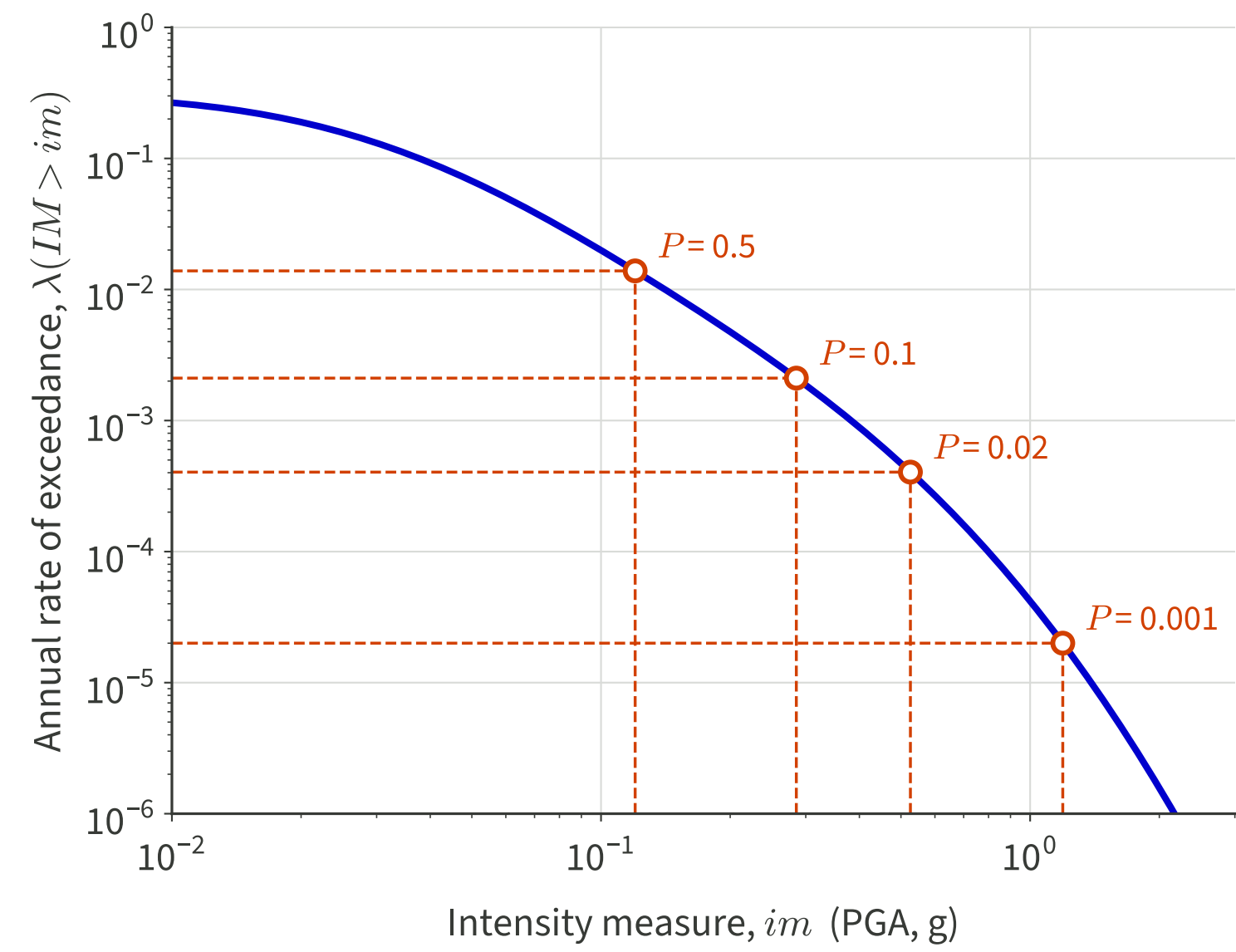
Fixed probability, varying design life



Probability of exceedance	Design life (yr)	$\lambda(IM > im^*)$ (/yr)	Return period (yr)	im^* (PGA, g)
0.1	1	0.105	9	0.04
0.1	50	2.11×10^{-3}	475	0.29
0.1	100	1.05×10^{-3}	949	0.38
0.1	10,000	1.05×10^{-5}	94,912	1.37

- The same 10% probability of exceedance requires a much higher design intensity for a longer design life
- Doubling T_{DL} halves the target rate, but the design intensity rises by much less than a factor of two: that depends on the *slope* of the hazard curve

Fixed design life, varying probability



Design life (yr)	Probability of exceedance	$\lambda(IM > im^*)$ (/yr)	Return period (yr)	im^* (PGA, g)
50	0.5	0.0139	72	0.12
50	0.1	2.11×10^{-3}	475	0.29
50	0.02	4.04×10^{-4}	2,475	0.53
50	0.001	2.00×10^{-5}	49,975	1.19

- For a 50-year design life, lowering the acceptable probability of exceedance from 10% to 2% moves the design motion from the 475-year to the 2,475-year level
- The tables and figures are computed from the same hazard curve: use the interactive figure to check any row

Typical return periods

- The return period is the reciprocal of the annual rate of exceedance:

$$RP = \frac{1}{\lambda(IM > im)}$$

- It is the average number of years *per* exceedance, but it is better not to think of it as a period of time: focus on the rate, “*in any given year, we expect this many exceedances*”
- Typical return periods that you will encounter in practice:
 - 475 years for ultimate limit states (10% probability in 50 years)
 - 2,475 years for collapse prevention (2% in 50 years)
 - 10^4 to 10^5 years for nuclear facilities, and longer for radioactive waste repositories

Computing the hazard curve

Annual rate of exceedance

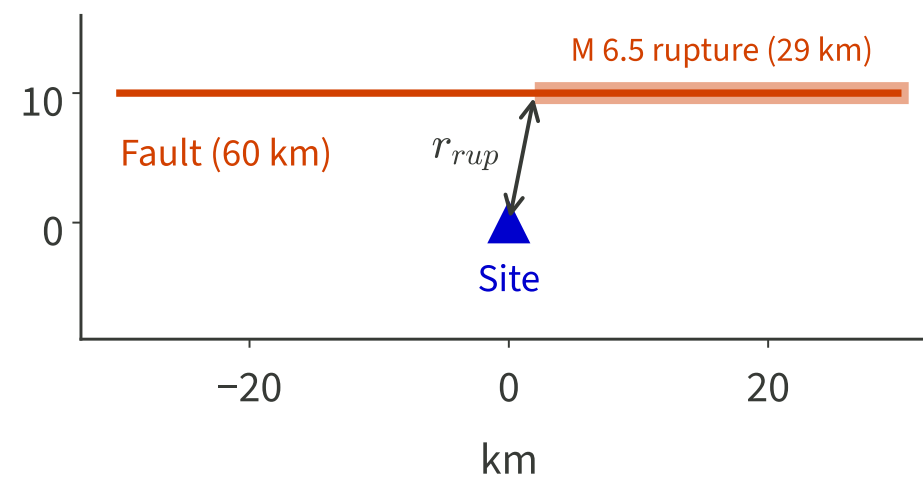
- Our key job *as engineering seismologists* is to determine the hazard curve, $\lambda(IM > im)$, for the IM that the problem requires
- It is obtained by summing, over all possible ruptures, the probability that each produces an exceedance, weighted by how often it occurs:

$$\lambda(IM > im) = \sum_{j=1}^{n_{sources}} \sum_{i=1}^{n_{rup,j}} P(IM > im \mid rup_{i,j}, site) \lambda_j(rup_{i,j})$$

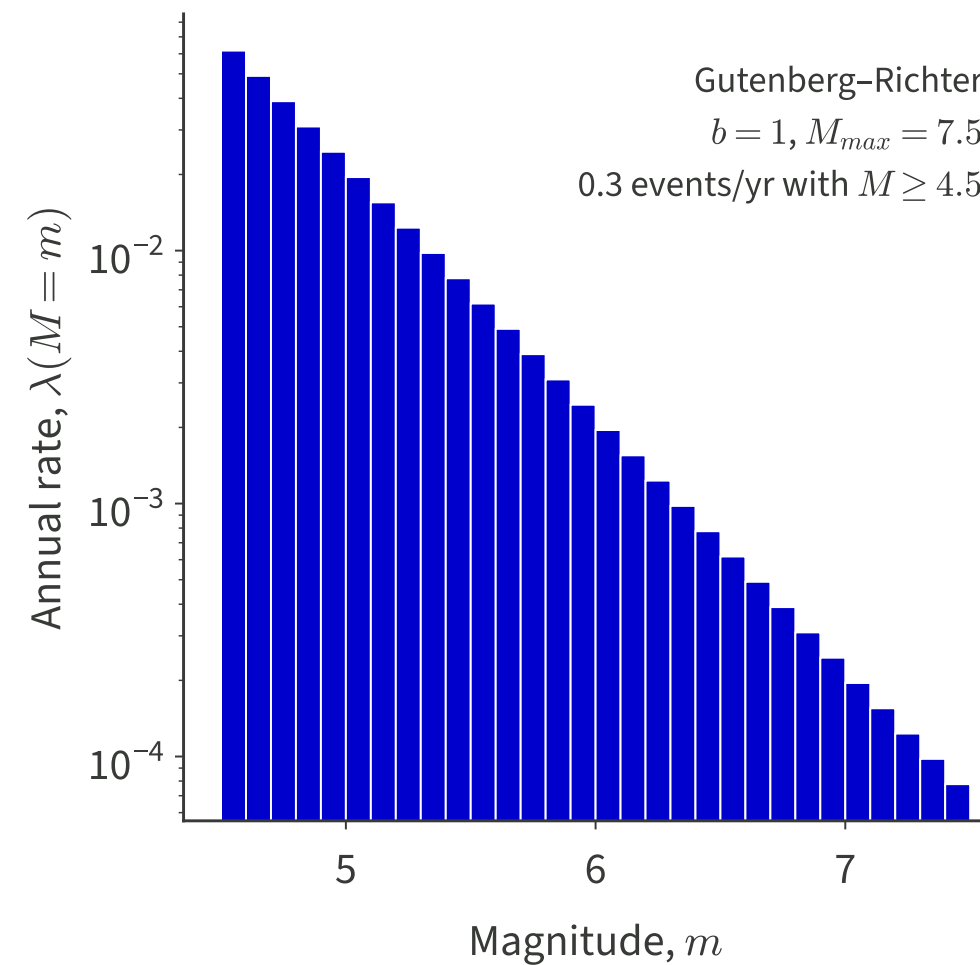
- To compute it we need:
 - a model of the seismic sources: where ruptures can occur [Lectures 2-3](#)
 - the rates of occurrence of rupture scenarios [Lectures 4-6](#)
 - the distribution of ground motion given a rupture [Lectures 7-9](#)
 - a probabilistic framework that combines them, with its uncertainties [Lectures 10-12](#)
- To go on to risk, the hazard curve is convolved with fragility or vulnerability functions, as in the introduction

Components of seismic hazard analysis

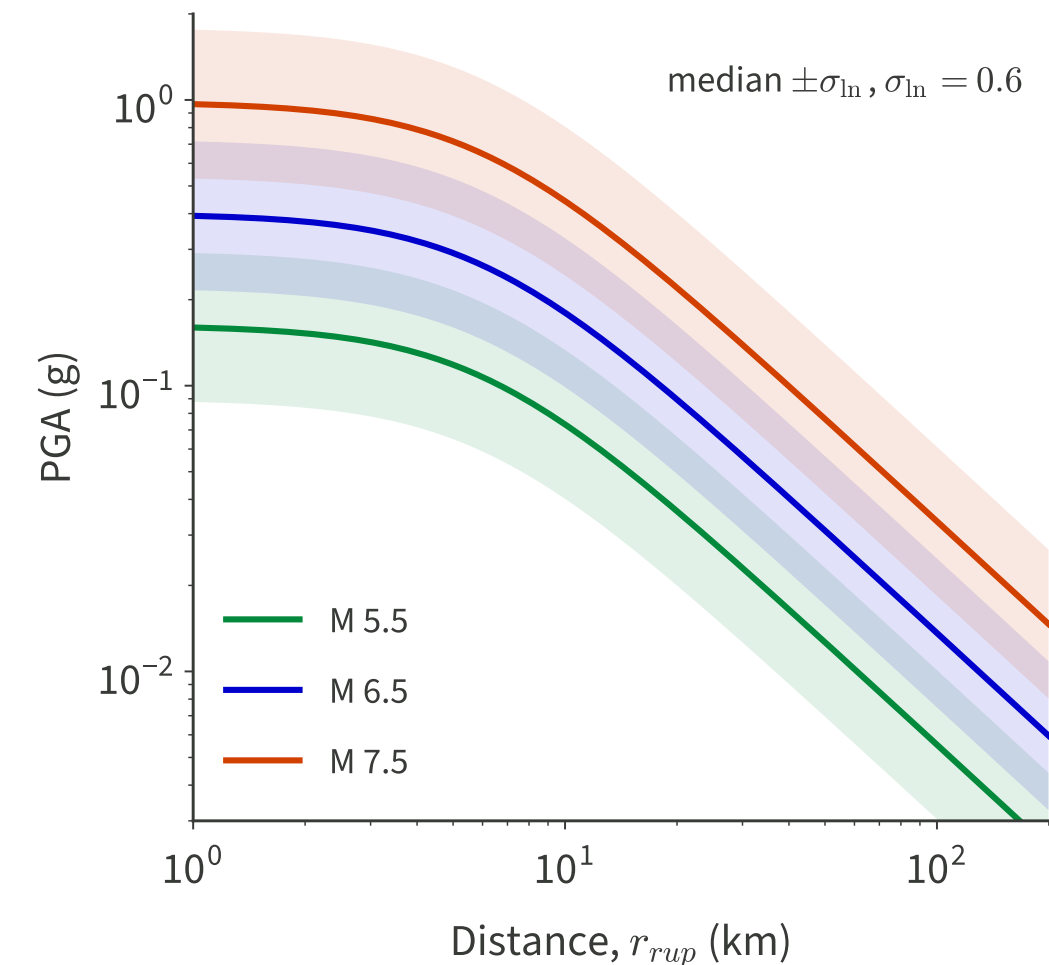
(a) Seismic source: where ruptures occur



(b) Recurrence: how often

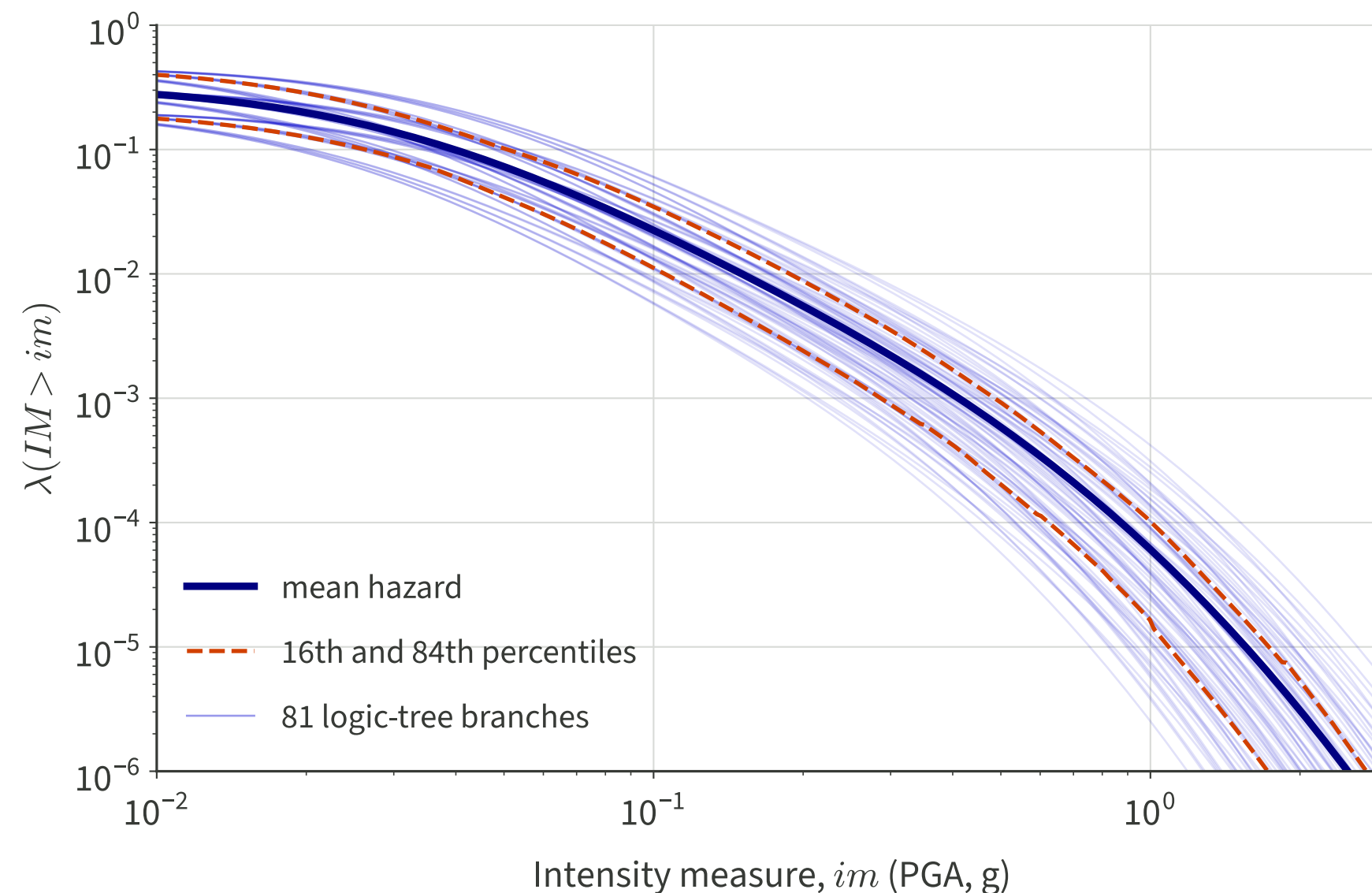


(c) Ground motion: how strong



- The toy PSHA behind this lecture: a 60 km fault 10 km from the site, Gutenberg–Richter recurrence, and a ground-motion model with lognormal scatter
- Every rupture on the fault, of every magnitude, contributes to the hazard curve through $P(IM > im \mid rup) \lambda(rup)$ [Lectures 2–9](#)

Uncertainty in the hazard curve



- Each model component is uncertain: the b -value, M_{max} , the activity rate and the ground-motion median
- A **logic tree** combines alternative models, each with a weight; every branch gives a hazard curve
- The spread of branch curves is the *epistemic* uncertainty; the mean hazard is normally used for design
- Here, 81 branches from three alternatives for each of four parameters [Lecture 12](#)

Exercise / Tutorial Problem

Tutorial problem

Consider two buildings, A and B , both with design lives of 50 years.

Building A is a generic office building and should be designed for a level of ground-motion that has a 10% chance of being exceeded over its life.

Building B is a critical facility for which the consequence of failure is very high. The probability of exceedance for the design level ground-motion for Building B should only be 1%.

- Compute the ratio of the return periods for buildings B and A .
- If the average annual rates of exceedance of ground-motion levels im is defined by:

$$\lambda(IM > im) = 0.05 \exp(-2im)$$

compute the corresponding ratio of design ground-motion levels for the two buildings.

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